

Attendance Calculator

Know where you stand before you skip.

System

79.4%

OVERALL ATTENDANCE

1 subject is below target. Check the red cards first.

Classes attended
158 / 199

Subjects
5

Safe
3

On the edge
1

Below target
1

Quick check

A one-off calculation. Nothing is saved.

Attended

Total held

Target %

42

50

75

84%

Safe

You can skip **6 classes** and stay at or above 75%.

What-if planner

See where you'll be after the next few classes.

Based on

Physics

Attend next

Miss next

5

1

68.09% → 69.81%

Below target against 75%. That's 37 of 53 classes.

Subjects

Mark each class as you go. Everything is saved in this browser.

Search subjects

Order added

Subject name

Attended

Total

Target %

e.g. Mathematics

0

0

Default

+ Add subject

84%

Mathematics

Safe

42 / 50 attended · target 75%

68%

Physics

Below target

32 / 47 attended · target 75%

75%

Chemistry

On the edge

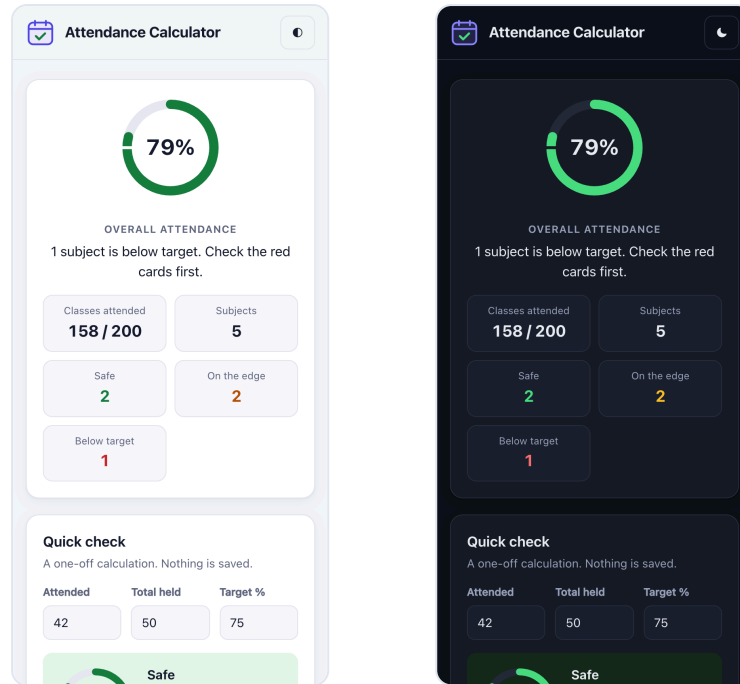
36 / 48 attended · target 75%

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Features

Area	What it does
Overview	Overall attendance across all subjects, a progress ring and counts of subjects that are <i>Safe</i> , <i>On the edge</i> and <i>Below target</i> .
Quick check	Enter attended, total and target %. You instantly see your percentage and how many classes you can skip or must attend. Nothing is saved.
What-if planner	"If I attend the next 5 and miss 1, where will I be?" Works from the Quick check numbers, one subject or all subjects combined.
Subject tracker	Add subjects with their existing counts, then mark each class Present or Absent with one click.
Per-subject targets	Each subject can have its own target (for example 80% for a lab) or use the default.
History	Every mark is saved with its date. Open a card's <i>History</i> to see recent marks and a colour strip of the last 30.
Undo	Undo the last mark on a subject. Deleting a subject, importing a backup or resetting everything can be undone from the pop-up message.
Edit	Fix a subject's name, counts or target at any time.
Search & sort	Filter subjects by name. Sort by name, lowest % first or highest % first.
Themes	System, Light or Dark. The choice is remembered.
Backup & restore	Export everything to a <code>.json</code> file and import it on another device or browser.
Spreadsheet export	Download a <code>.csv</code> with each subject's numbers, status and advice.
Printable report	Print or <i>Save as PDF</i> a clean report of the overview and subject cards.
Responsive	Works on phones, tablets and desktops. Supports keyboard use and screen readers.



Getting started

Run it

1. Download or copy the project folder.
2. Double-click `index.html` . It opens in your default browser.

On macOS you can also run this from the project folder:

```
open index.html
```

No build step, package manager or internet connection is needed.

Optional: serve it locally

If you'd rather use a local web server (for example to open it from your phone on the same Wi-Fi):

```
python3 -m http.server 8000
# then visit http://localhost:8000
```

Note: Browsers keep saved data separately for each address. Data saved while using `file:///.../index.html` won't appear at `http://localhost:8000` , and the reverse is also true. Use **Export backup** and **Import backup** to move it.

Supported browsers

Current versions of Chrome, Edge, Firefox and Safari (Safari 16.2 or later) on desktop and mobile.

How to use it

1. Quick check

Use this for a one-off answer without saving anything.

1. Enter **Attended** (classes you were present for).
2. Enter **Total held** (all classes held so far).
3. Check **Target %** (it starts at your default target, usually 75%).

The result box turns green, amber or red and tells you exactly what to do. For example, 42 of 50 classes gives **84%**, and you *can skip 6 classes and stay at or above 75%*.

2. What-if planner

1. Choose what to base it on under **Based on:** *Quick check numbers*, *All subjects combined* or a single subject.
2. Enter how many classes you'll **attend next** and **miss next**.

It shows your percentage now and afterwards (for example *68.09% → 69.81%*) and whether that meets the target.

3. Tracking subjects

Add a subject. Type a name. You can also enter the classes you've already attended and held, and a custom target. Leave the target blank to use the default. Then click **+ Add subject**.

Mark classes. On a subject card:

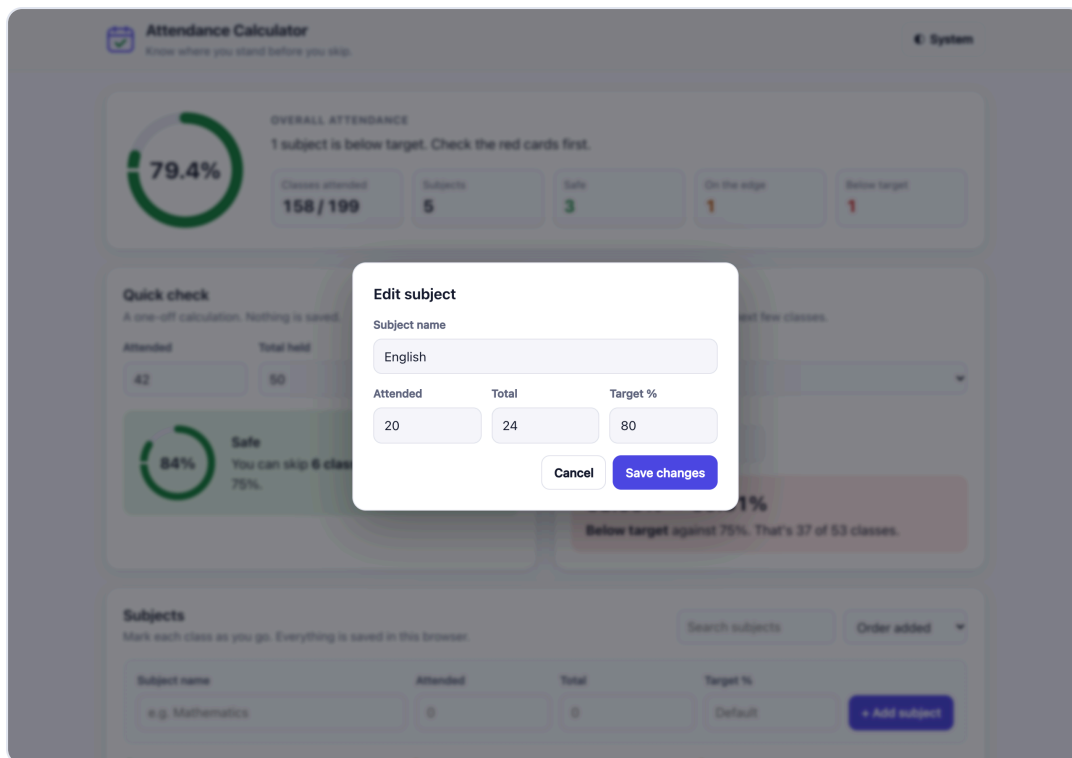
- **✓ Present** adds one attended class and one held class.
- **✗ Absent** adds one held class.
- **Undo** reverses the most recent mark on that subject.

Read the card. Each card shows:

- a ring and a bar with your percentage (the small mark on the bar is your target),
- a status badge (see the table below),
- one line of advice, such as *"Attend the next 13 classes in a row to reach 75%",*
- **History**, which lists each mark with its date.

Edit or delete. Use the  and  icons at the top right of a card. After a delete, click **Undo** in the pop-up message within 6 seconds to bring the subject back.

Find subjects. Use the search box, or change the sort order to *Lowest % first* to see which subjects need attention.



4. What the statuses mean

Status	Colour	Meaning
Safe	Green	At or above target, and you can skip at least one class.
On the edge	Amber	At or above target, but skipping the next class would drop you below it.
Below target	Red	Under target. The advice tells you how many classes to attend in a row.
No classes yet	Grey	No classes have been marked for this subject.

5. Settings & data

Control	What it does
Default target %	Target used by every subject without its own target, by the overview and by the Quick check.
Theme button (top right)	Switches between <i>System</i> , <i>Light</i> and <i>Dark</i> .
Export backup (.json)	Downloads all subjects, history and settings.
Import backup	Loads a backup file and replaces the current data. It asks you first and can be undone.
Export spreadsheet (.csv)	Downloads a table that opens in Excel, Google Sheets or Numbers.
Print / save report	Opens the print dialog. Choose <i>Save as PDF</i> to keep a copy.
Reset everything	Deletes all subjects and history, but keeps your settings. It asks you first and can be undone.

How the calculations work

Let:

- **A** = classes attended
- **T** = total classes held
- **P** = target percentage (1–100)

Current percentage

$$\text{percentage} = A \div T \times 100$$

Classes you must attend (when below target)

This is the smallest number **x** of consecutive classes you must attend so that $(A + x) \div (T + x) \geq P \div 100$:

$$x = \text{ceil}((P \times T - 100 \times A) \div (100 - P))$$

If the target is **100%** and you've already missed a class, the target can't be reached, and the app says so.

Classes you can skip (when at or above target)

This is the largest number **y** of classes you can miss so that $A \div (T + y) \geq P \div 100$:

$$y = \text{floor}((100 \times A - P \times T) \div P)$$

What-if projection

$$\text{new percentage} = (A + \text{attendNext}) \div (T + \text{attendNext} + \text{missNext}) \times 100$$

All comparisons use whole-number arithmetic (for example $A \times 100 < P \times T$ rather than dividing first), so results such as "exactly 75%" aren't affected by floating-point rounding.

Worked examples (target 75%)

Attended / Total	Percentage	Result	Check
42 / 50	84%	Can skip 6	$42 \div 56 = 75\%$ ✓
30 / 50	60%	Must attend 30	$60 \div 80 = 75\%$ ✓
36 / 48	75%	On the edge (skip 0)	$36 \div 49 = 73.5\%$ ✗
32 / 47	68.09%	Must attend 13	$45 \div 60 = 75\%$ ✓

Your data and privacy

- All data is stored in your browser's `localStorage` under the key `attendance-app-v2`.
- Nothing is sent over the network. The app makes no network requests at all.
- Data is kept separately for each **browser, device and address**. Clearing site data, or using a private window, removes it. Export a backup regularly if the data matters to you.
- Data from version 1 of the app (key `attendance-subjects`) is imported automatically the first time version 2 opens.

Backup file format

```
{
  "version": 2,
  "settings": { "target": 75, "theme": "system" },
  "subjects": [
    {
      "id": "m1abc2de3f",
      "name": "Physics",
      "attended": 32,
      "total": 47,
      "target": null,
      "log": [
        { "t": "p", "d": "2026-09-23" },
        { "t": "a", "d": "2026-09-24" }
      ]
    }
  ]
}
```

Field	Meaning
<code>target</code>	The subject's own target, or <code>null</code> to use the default.
<code>log[].t</code>	"p" = present, "a" = absent.
<code>log[].d</code>	Date of the mark (YYYY-MM-DD , local time).

Imported files are checked and cleaned before use. Invalid counts are clamped, attended can never exceed total, names are limited to 40 characters and unknown fields are ignored.

Project structure

```
attandacne/
├─ index.html      Page structure and the edit dialog
├─ style.css       Colours, light/dark themes, layout and print styles
├─ app.js          Calculations, saving, rendering and import/export
├─ README.md       This documentation
└─ docs/
    ├─ Attendance-Calculator-Documentation.pdf
    └─ images/      Screenshots used in the documentation
```

For developers

Architecture

The app is plain HTML, CSS and JavaScript with no frameworks or dependencies. `app.js` is organised in sections:

Section	Responsibility
Core math	Pure functions with no DOM access: <code>analyze()</code> , <code>project()</code> , <code>statusOf()</code> and <code>adviseHtml()</code> .
Helpers	Input parsing (<code>readInt</code>), HTML escaping (<code>esc</code>), dates, the SVG ring renderer and file downloads.
State & persistence	<code>state</code> object, <code>load()</code> , <code>save()</code> and <code>normalize()</code> , which validates saved and imported data and migrates version 1.
Toast	Pop-up messages with an optional <i>Undo</i> action.
Theme	Cycles <code>system</code> → <code>light</code> → <code>dark</code> by setting <code>data-theme</code> on <code><html></code> .
Overview / Quick check / What-if / Subjects	Each has a <code>render...()</code> function that rebuilds its part of the page from <code>state</code> .
Settings & data	Default target, JSON export and import, CSV export, print and reset.

Data flow. Every user action changes `state` and then calls `commit()`, which saves to `localStorage` and re-renders everything. The page is always drawn from `state`, so what you see can't drift out of sync with what's saved.

Key functions

```
analyze(attended, total, target)    // → { percent, needed, canSkip }
project(attended, total, attend, miss) // → { attended, total, percent }
statusOf(result, target, total)     // → { key: "good"|"warn"|"bad"|"new", label }
normalize(rawJson)                  // → clean state, or throws on unrecognised input
```

Security notes

- User-entered names are always escaped with `esc()` before being inserted into HTML.
- CSV cells that start with `=`, `+`, `-` or `@` get a leading `'` so spreadsheets don't run them as formulas.
- Imported JSON is never trusted as-is. `normalize()` rebuilds every field.

Theming

Colours are CSS custom properties on `:root` in `style.css`. Dark values apply in two cases: under `prefers-color-scheme: dark` (unless Light is chosen), and when `data-theme="dark"` is set. To change the palette, edit the token blocks at the top of the file.

Testing

The app was tested in headless Google Chrome by driving the real page. Each of these scenarios was checked:

- empty state, adding subjects, rejecting duplicate names and attended values greater than total,
- Present / Absent / Undo counts and the advice text,
- overview totals and status counts,
- Quick check results and input errors,
- What-if projection for a single subject,
- search and sort order,
- data surviving a page reload,
- editing a subject, and deleting it and bringing it back with Undo,
- dark theme, and a phone-width layout with no horizontal scrolling,
- no console errors.

The calculation formulas were also checked against the worked examples above.

Troubleshooting

Problem	Fix
My subjects disappeared.	You may be in a different browser, a private window, or opening the page from a different address (<code>file://</code> vs <code>http://</code>). Import your latest backup.
Import says "doesn't look like an attendance backup".	The file must be a <code>.json</code> exported by this app (or have a <code>subjects</code> array).
The theme doesn't change.	<i>System</i> follows your device setting. Click the theme button again to pick <i>Light</i> or <i>Dark</i> .
The printed report has no colours.	Turn on Background graphics in the print dialog's options.
A mark was added by mistake.	Click Undo on that subject's card.

Limitations and ideas

Current limitations

- Data isn't synced between devices automatically. Use export and import.
- Marks are dated *today*. You can't choose a past date, but you can adjust the counts with **Edit**.
- There's no timetable, so each class has to be marked by hand.

Possible next steps

- Weekly timetable with "mark today's classes" in one tap.
- Choosing a date for each mark, and a calendar view.
- Installable offline app (PWA).
- Charts showing how attendance has changed over time.

Attendance Calculator. Version 2.0, September 2026.